CROWD MANAGEMENT AND PILGRIM SAFETY

The Simhasth will involve the movement and congregation of very large numbers of people across bathing ghats, temples, religious venues, transit points, parking areas, marketplaces, accommodation zones, processional routes and other public spaces. Crowd conditions can change rapidly. A location that is operating normally can experience a sudden increase in density because of an event, route closure, delayed movement, weather conditions, transport arrivals or changes in pilgrim behaviour. The core challenge is therefore not merely counting people, but understanding crowd dynamics, identifying emerging risks and enabling timely intervention while preserving privacy, dignity and the religious character of the event. Authorities need reliable situational awareness across dispersed locations and the ability to coordinate field teams based on a common operating picture.

Some of the key challenges include:

- Dynamically changing crowd density, quick crowd build-up and volatile movement patterns.
- Large volumes of pedestrian movement at ghats, narrow corridors, entrance-exit points - especially during major religious events
- Counterflow and conflicting pedestrian movement. Queue management at high demand corridors.
- Risks related to separated and lost people amid large crowds
- Effectively communicating information to pilgrims from different parts of the country about route changes and safety advisories
- Managing crowd safety and movement for different conditions like heat stress, heavy rainfall, high decibel levels, etc.
- Accessibility and mobility limitations for vulnerable populations like elderly citizens, children, people with disabilities, etc.

Participants are expected to exhibit capabilities in technology, urban design & planning, and/or behavioural science. Proposals should address the aforementioned challenges and potential areas of applications can include, but are not limited to:

- Real-time analytics
- Predictive modelling and simulation
- Decision-support systems
- Digital tools for pilgrims
- Accessibility improvement to existing built environment
- Effective mass communication strategies

This is the general direction in which applicants are encouraged to think. However, applicants are welcome to submit proposals rooted in principles that do not feature in this list but are capable of address challenges of crowd management and pilgrim safety anticipated during the Simhasth.